

DECIMAL ASCII CODES

FOR FIN<> AND FOUT<>

CODE CHARACTER	CODE CHARACTER	CODE CHARACTER
128 CTRL/SHIFT/P (LEADER)	160 SPACE	193 A
129 CTRL/A	161 !	194 B
130 CTRL/B	162 "	195 C
131 CTRL/C	163 #	196 D
132 CTRL/D	164 \$	197 E
133 CTRL/E	165 %	198 F
134 CTRL/F	166 &	199 G
135 CTRL/G (BELL)	167 '	200 H
136 CTRL/H	168 (	201 I
137 CTRL/I	169)	202 J
138 LINE FEED	170 *	203 K
139 CTRL/K	171 +	204 L
140 CTRL/L	172 ,	205 M
141 CARRIAGE RETURN	173 -	206 N
142 CTRL/N	174 .	207 O
143 CTRL/O	175 /	208 P
144 CTRL/P	176 0	209 Q
145 CTRL/R	177 1	210 R
146 CTRL/S	178 2	211 S
147 CTRL/T	179 3	212 T
148 CTRL/U	180 4	213 U
149 CTRL/V	181 5	214 V
150 CTRL/W	182 6	215 W
151 CTRL/X	183 7	216 X
152 CTRL/Y	184 8	217 Y
153 CTRL/Z	185 9	218 Z
154 CTRL/SHIFT/K	186 :	219 [
155 CTRL/SHIFT/L	187 ;	220 \
156 CTRL/SHIFT/M	188 <	221]
157 CTRL/SHIFT/N	189 =	222 ^
158 CTRL/SHIFT/O	190 ?	223 _
159 CTRL/SHIFT/P	191 ?	253 ALT MODE
	192 @	255 RUBOUT

FOUT(141) WILL OUTPUT A RETURN/LINE FEED; FOUT(13) WILL OUTPUT A CARRIAGE RETURN ONLY. CODE 144 (CTRL/P) IS PS/8 FOCAL'S BREAK CHARACTER. 224 THROUGH 250 ARE LOWER-CASE LETTERS ON SOME TERMINALS. MANY TERMINALS USE SHIFT/K, /L, AND /M FOR '[', '\', AND ']'.

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PS/8 FOCAL, 1971 ERROR CODES

ERROR CODES VALID ONLY FOR 6/1/71 VERSION OF PS/8 FOCAL

CODE EXPLANATION

700.00	MANUAL START FROM CONSOLE
701.00	BREAK FROM KEYBOARD VIA 'CTRL/P'
701.40	BAD LINE NUMBER
701.91	GROUP NUMBER LARGER THAN 31
701.13	ZERO IS ILLEGAL LINE NUMBER
702.32	'DO' REFERENCED NON-EXISTENT GROUP
702.52	'DO' REFERENCED NON-EXISTENT LINE
703.05	NON-EXISTENT LINE AFTER 'GOTO' OR 'IF'
703.28	ILLEGAL COMMAND
704.39	BAD VARIABLE IN 'FOR' OR 'SET' COMMAND
704.52	EXCESS RIGHT TERMINATORS
704.60	ILLEGAL TERMINATOR IN 'FOR' OR 'SET' COMMAND
704.13	MISSING ARGUMENT IN DISPLAY FUNCTION
705.48	BAD ARGUMENT IN 'MODIFY' COMMAND
706.06	ILLEGAL USE OF A FUNCTION OR NUMBER
706.54	TOO MANY VARIABLES
707.38	NO OPERATOR BEFORE PARENTHESIS
707.19	FUNCTION ARGUMENT MISSING
707.16	ILLEGAL FUNCTION OR DOUBLE OPERATORS
708.47	ENCLOSURES DO NOT MATCH
709.11	BAD ARGUMENT IN 'ERASE' COMMAND
710.15	STORAGE FILLED BY TEXT
711.35	INPUT BUFFER OVERFLOW
715.28	BAD RESTORE COMMAND
715.11	ATTEMPT TO WRITE PAST END-OF-FILE
720.34	LOGARITHM OF ZERO REQUESTED
722.23	NO OUTPUT FILE TO RESTORE
722.51	BAD FILES 'OPEN' COMMAND
723.05	DATA FILE NOT FOUND
723.15	NO INPUT FILE TO RESTORE
723.36	TOO MANY DIGITS IN NUMBER
723.39	PUSHDOWN OVERFLOW; PROGRAM TOO LONG
724.60	TWO PERIODS IN FILE NAME OR NO NAME
725.37	BAD DEVICE OR 2 PAGE HANDLER
726.18	BAD LIBRARY COMMAND
726.58	NO PROGRAM NAMED BY 'SAVE' COMMAND
726.99	EXPONENT TOO LARGE OR NEGATIVE
727.10	PROGRAM NOT FOUND
727.14	'LIBRARY CALL' FROM NON-DIRECTORY DEVICE
728.06	'LIBRARY LIST' OF NON-DIRECTORY DEVICE
728.73	DIVISION BY ZERO
729.53	FILE ALREADY DELETED
729.68	DEVICE ERROR: WRITE LOCK OR PARITY
729.91	CANNOT OPEN OUTPUT FILE
730.05	IMAGINARY SQUARE ROOTS REQUIRED
731.47	UNAVAILABLE COMMAND OR FUNCTION
?	ATTEMPT TO READ PAST END-OF-FILE (INPUT IS SWITCHED TO TERMINAL AND PROGRAM CONTINUES)

TYPESET ON CENTRONICS-101 MATRIX PRINTER

ADDITIONAL COPIES OF SUMMARY CARD AVAILABLE FROM:

OMSI SOFTWARE DEVELOPMENT GROUP
4015 SW CANYON ROAD
PORTLAND, OREGON 97221

PRINTED IN U.S.A. 7/28/71



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PS/8 FOCAL

SUMMARY CARD

ELEMENTARY COMMANDS

*ASK <X> ACCEPT VALUE OF X FROM INPUT DEVICE
*COMMENT IGNORE REST OF LINE
*DO [LINE OR GROUP NUMBER] SUBROUTINE CALL
*ERASE ZERO VARIABLES
*ERASE <LINE OR GROUP NUMBER> DELETE LINE OR GROUP
*ERASE ALL DELETE ENTIRE INDIRECT PROGRAM
*FOR X=E1,E2,E3, COMMAND REPEATED (E3-E1)/E2+1 TIMES
*GO [L1] START PROGRAM EXECUTION AT LINE L1
*IF (E1),L2,L3 BRANCH ON SIGN OF E1
*MODIFY <L1> EDIT LINE L1
*QUIT STOP EXECUTION
*RETURN TERMINATE 'DO' OR 'GOSUB' SUBROUTINE
*SET X=<E1> ASSIGN VALUE E1 TO VARIABLE
*TYPE [NUMBERS, E1, "TEXT", #, #, # OR X] OUTPUT
*WRITE [LINE OR GROUP NUMBER] LIST PROGRAM

LIBRARY AND FILE COMMANDS

*LIBRARY CALL [DEVICE:]<PROGRAM> LOAD PROGRAM
*LIBRARY DELETE [DEVICE:]<PROGRAM> UNSAVE PROGRAM
*LIBRARY GOSUB [DEVICE:]<PROGRAM> [G1] SUBROUTINE CALL
*LIBRARY LIST [DEVICE:]<FIRST LISTED FILE> CATALOG
*LIBRARY RUN [DEVICE:]<PROGRAM> [L1] CHAIN
*LIBRARY SAVE [DEVICE:]<PROGRAM> SAVE PROGRAM
*LIBRARY EXIT RETURN TO PS/8 MONITOR

*OPEN INPUT [DEVICE:]<FILE>[,ECHO] PREPARE TO READ
*OPEN OUTPUT [DEVICE:]<FILE>[,ECHO] PREPARE TO WRITE
*OPEN RESTORE INPUT[,ECHO] RESUME INPUT FROM FILE
*OPEN RESTORE OUTPUT[,ECHO] RESUME OUTPUT TO FILE
*OUTPUT CLOSE OUTPUTS BUFFER; CLOSING OUTPUT FILE

---INPUT---,ECHO ECHO INPUT ONTO OUTPUT DEVICE
---OUTPUT---,ECHO ECHO OUTPUT ONTO TERMINAL
FC & FD ARE THE PROGRAM & DATA FILE NAME EXTENSIONS.

<> ENCLOSE REQUIRED TERMS. [] ENCLOSE OPTIONAL TERMS.
ONE LETTER ABBREVIATIONS MAY BE USED AS COMMAND WORDS.
X REPRESENTS A VARIABLE. E1, E2 AND E3 ARE ARITHMETIC EXPRESSIONS. L1, L2 AND L3 ARE LINE NUMBERS. G1 IS A LINE OR GROUP NUMBER. L1-L3 AND G1 CAN BE REPLACED BY ARITHMETIC EXPRESSIONS WHICH DO NOT BEGIN WITH AN 'A'.

SYMBOLS AND CHARACTERS

* / + - ... EXPONENT, MULTIPLY, DIVIDE, ADD, SUBTRACT
 () [] < > ... ENCLOSURES FOR ARITHMETIC EXPRESSIONS
 ! TYPE '!' OR 'ASK !' OUTPUTS A RETURN/LINE FEED
 # TYPE '#' OR 'ASK #' OUTPUTS A RETURN ONLY
 " TYPE "STRING" OR 'ASK "STRING"' OUTPUTS 'STRING'
 % TYPE '%26.*' TABS TO LOCATION 26 AND TYPES '*'
 \$ TYPE '\$' OUTPUTS SYMBOL TABLE
 % TYPE '%5.02' FORMATS OUTPUT (5 DIGITS, XXX.XX)
 ? GO ? TRACES PROGRAM UNTIL NEXT '?'
 E POWERS-OF-TEN NOTATION (5.3E+06 EQUALS 5300000)
 , SEPARATES EXPRESSIONS IN 'TYPE' AND 'ASK' COMMANDS
 ; INSERTED BETWEEN COMMANDS; E.G. 'DO X;TYPE Y/7.2'
 CTRL/Z END-OF-FILE CHARACTER
 COMMA SPACE OR RETURN ENDS NUMBER BEING 'ASKED'
 ALT MODE IN RESPONSE TO 'ASK X' LEAVES OLD VALUE OF X
 CTRL/O CHANGES SEARCH CHARACTER IN 'MODIFY'
 CTRL/L SKIPS TO NEXT OCCURRENCE OF SEARCH CHARACTER
 LINE FEED RETAINS REST OF LINE IN 'MODIFY'
 RETURN KILLS REST OF LINE IN 'MODIFY'
 RUBOUT DELETES PRECEDING CHARACTER IN LINE
 ← KILLS PROGRAM LINE BEING ENTERED
 ← KILLS NUMBER BEING ENTERED IN RESPONSE TO 'ASK'
 CTRL/P PS/8 FOCAL'S BREAK CHARACTER

PS/8 DEVICE NAMES

SYS: SYSTEM DEVICE (DISK OR DECTAPE DRIVE #0)
 DSK: DISK OR DECTAPE DRIVE #0; HOWEVER, DECTAPE
 INSTALLATIONS SHOULD ASSIGN DTA1: DSK:
 'DSK:' IS THE DEVICE IF NONE IS SPECIFIED.
 DTA0: - DTA7: DECTAPE DRIVES
 LTA0: - LTA7: LINTAPE DRIVES
 MTA0: - MTA7: MAGTAPE DRIVES
 PTR: HIGH SPEED READER
 PTP: HIGH SPEED PUNCH
 LPT: LINE PRINTER
 TTY: TERMINAL

PROTECTED VARIABLES

! , ' ' AND ' ' CANNOT BE 'TYPED', 'ASKED' OR 'ERASED'
 BUT ARE OTHERWISE NORMAL VARIABLES. '0' IS THE SYSTEM
 STATUS VARIABLE AND IS 'SET' TO THE PRODUCT OF CODES
 REPRESENTING FEATURES CHOSEN IN THE INITIAL DIALOGUE.

FEATURE	CODE
'0' NO OPTIONAL FEATURES	1
'1' FSIN AND FCOS FUNCTIONS	2
'2' FSIN, FCOS, FLOG, FEXP AND FATN	6
'3' EXTENDED PRECISION (10 DIGITS)	5
'4' FILE COMMANDS	7
'5' LEADING SPACE IN 'TYPE'	11
'6' ' ' AUTOMATICALLY PRINTED BY 'TYPE'	13
'7' ' ' AUTOMATICALLY PRINTED BY 'ASK'	17
'8' 'E' IN POWERS-OF-TEN NOTATION	-1
'9' FEATURES 2,3,4	210
'YES' STANDARD FEATURES 2,3,4,5	2310

FUNCTIONS

FSIN(A) SINE (RADIANS) FITR(A) INTEGER PART
 FCOS(A) COSINE (RADIANS) FSN(A) SIGN PART (-1, 0, +1)
 FATN(A) ARCTAN (RADIANS) FDIS(X,Y) 340 SCOPE DISPLAY
 FEXP(A) EXPONENTIAL FADC(A) ANALOG TO DIGITAL
 FLOG(A) LOG (BASE E) FRAN(X) PSEUDO-RANDOM NUMBER
 FSQRT(A) SQUARE ROOT FIN(X) CHARACTER INPUT
 FABS(A) ABSOLUTE VALUE FOUT(A) CHARACTER OUTPUT

DISPLAY & JOYSTICK FUNCTIONS FOR THE TEKTRONIX T-4002 GRAPHIC COMPUTER TERMINAL

FDIS(I,X,Y) --- DISPLAY

COORDINATES: 0 ≤ X < 1024 0 ≤ Y < 1024

I=-1 PLOT A POINT AT (X,Y)
 I=0 DRAW A DARK VECTOR TO (X,Y)
 I=+1 IF IN GRAPHICS MODE, DRAW A BRIGHT VECTOR TO
 (X,Y); OTHERWISE ENTER GRAPHICS MODE AND DRAW
 A DARK VECTOR TO (X,Y).

FJOY(I) ----- JOYSTICK

FJOY(I) SETS VARIABLES XJ AND YJ EQUAL TO THE X AND Y
 LOCATION ADDRESSES IN THE INTERACTIVE GRAPHICS UNIT.

I=-1 SETS XJ AND YJ EQUAL TO LOCATIONAL ADDRESSES.
 I=0 TURNS ON JOYSTICK. WAITS FOR KEYBOARD
 INPUT BEFORE READING CURSOR COORDINATES.
 I=+1 ASSUMES JOYSTICK PREVIOUSLY TURNED ON BY
 A FOUT(17); READS CURSOR IMMEDIATELY.

GRAPHICS FOUT(< >) CODES:

1 ... HOME
 17 ... TURNS ON JOYSTICK FOR FJOY(1) APPLICATION
 24 ... ERASE SCREEN (REQUIRES .5 SECOND DELAY)
 30 ... ENTER INCREMENTAL PLOT MODE
 31 ... LEAVE GRAPHICS MODE

FOUT(< >) CODES FOR INCREMENTAL PLOT MODE:

DIRECTION	N	NE	E	SE	S	SW	W	NW
BRIGHT POINT	32	33	34	35	36	37	38	39
DARK POINT	40	41	42	43	44	45	46	47

PROGRAM: 'SKETCH'

FOLLOWS JOYSTICK ON DISPLAY

```
01.10 SET I=0; FOR K=0,2,FOUT(24); C ERASE AND PAUSE
01.20 SET DUMMY=FJOY(0); C WAITS FOR A KEY TO BE HIT
01.30 SET DUMMY=FDIS(0,XHOLD,YHOLD)+FDIS(I,XJ,YJ)
01.40 SET XHOLD=XJ; SET YHOLD=YJ; SET I=1; GOTO 1.2
```

PROGRAMMING EXAMPLES

INPUT PROGRAM TAPE:

```
1) TYPE 'OPEN INPUT PTR;' (WAIT FOR '~' ON TERMINAL)
2) HIT SPACE BAR WHEN READY TO READ TAPE
3) WAIT FOR OUTPUT OF '?' ON TERMINAL; HIT 'CTRL/P'
```

TO INPUT PROGRAMS FROM THE TERMINAL'S LOW SPEED READER:

```
1) TYPE 'OPEN INPUT TTY;' (TURNS ECHO OFF)
2) READ IN TAPE
3) TYPE 'OPEN INPUT TTY;;ECHO' (TURNS ECHO ON)
```

INPUT DATA FROM READER:

```
10.60 OPEN INPUT PTR;
      (WAIT FOR '~' ON TERMINAL; HIT SPACE BAR)
10.65 FOR X=1,10;ASK A(X)
10.70 OPEN INPUT TTY;;ECHO; COMMENT RESTORES TERMINAL
```

OUTPUT PROGRAM TAPE:

```
1) TYPE 'OPEN INPUT TTY;;OPEN OUTPUT PTP;;WRITE ALL'
   (WAIT FOR PUNCH TO STOP)
2) TYPE 'OUTPUT CLOSE;OPEN INPUT TTY;;ECHO'
```

OUTPUT DATA TO PUNCH:

```
10.80 OPEN OUTPUT PTP;
10.85 FOR X=1,10;TYPE A(X)
10.90 OUTPUT CLOSE
```

LIST PROGRAM ON LINE PRINTER:

```
1) TYPE 'OPEN INPUT TTY;;OPEN OUTPUT LPT;;WRITE ALL'
2) 'OUTPUT CLOSE;OPEN INPUT TTY;;ECHO'
```

CHARACTERS AND FILES:

GROUP 1 SETS UP THE 'WORDS' FILE. GROUP 2 SCANS IT,
 ECHOING ONLY WORDS STARTING WITH THE SEARCH LETTER.

```
01.10 OPEN OUTPUT DTA2;WORDS; OPEN OUTPUT TTY;;ERASE
01.20 TYPE "ENTER WORDS, FOLLOW EACH WITH A SPACE:"
01.30 TYPE "<ENTER A SPACE BY ITSELF TO CLOSE FILE"!!
01.40 TYPE "5?" "OPEN RESTORE OUTPUT,ECHO
01.50 IF (-FABS(FIN(-160))>1.5;OPEN OUTPUT TTY;
01.60 TYPE "MORE?" "OPEN RESTORE OUTPUT,ECHO
01.70 IF (-FABS(FIN(-160))>1.5;OUTPUT CLOSE;GOTO 2.7

02.10 TYPE "ENTER SEARCH LETTER: " ;SET L=FIN(0);T !!!
02.20 TYPE "WORDS STARTING WITH " ;S 0=FOUT(L);T "!!";S
02.30 OPEN INPUT DTA2;WORDS;SET S=160;SET P=2.6
02.40 SET C=FIN(0);IF (FABS(C-S)>2.7);1 (-FABS(L-C)>2.6
02.50 OPEN RESTORE INPUT,ECHO;T ;5;SET DUMMY=FOUT(C)
02.60 IF (-FABS(FIN(-S))>P);OPEN RESTORE INPUT;G 2.4
02.70 OPEN INPUT TTY;;ECHO;TYPE "!!DONE "!!;QUIT
```